



Animated Ambient Logo Background — Design Specification

The real mark, extruded in true 3D, turning site-wide behind the content

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1 Overview

Replace the per-section static backdrops with a single **site-wide animated ambient background**: the **real NurvexThink mark, extruded into true 3D**, slowly turning like a physical object on a turntable, floating over slow-moving aurora glows and a faint blueprint grid. The watermark *is* the hero visual — the home headline sits directly over it, so the site has exactly **one** live WebGL scene.

The effect is **theme-aware** (opacity/lift per theme; the logo keeps its true colours in light mode) and reacts subtly to the mouse for depth. It renders **once**, behind all public pages, and is suppressed on `/admin`.

1.1 Goals

- A premium, branded atmosphere that reinforces the real mark on every public page.
- True 3D: depth, beveled edges, metallic shine as the mark turns.
- Smooth, GPU-friendly motion that never harms readability or Core Web Vitals.
- Fully theme-aware, honoring `prefers-reduced-motion`.

1.2 Non-goals

- No second live 3D scene (the old hero crystal/canvas was removed in favour of this one).
- No per-page configuration or CMS control of the background.
- Not shown on admin pages (kept clean and functional).

2 Locked decisions (as built)

Decision	Choice
Scope	Site-wide — one fixed layer behind all public pages; excluded on <code>/admin*</code>
Watermark	The real logo in true 3D — auto-traced outlines, extruded, beveled, metallic PBR
Motion	Slow full 360° turn (~22 s) + gentle float, plus mouse parallax on the layer (parallax off on touch and reduced-motion)
Colour	True logo colours in both themes (no light-mode tint); dark mode gets a brightness lift to read on black
Materials	Navy N: deep #1C2848, glossy clearcoat. Silver T: bright #DDE2EA, high metalness
Architecture	One WebGL scene site-wide (lazy-loaded R3F canvas, $dpr \leq 1.5$) + CSS-only aurora/grid

3 Approach

The watermark is a real extruded mesh (React Three Fiber), because a flat image cannot show true perspective, bevels, or specular shine. Everything else (aurora, grid, parallax) stays **pure CSS transforms/opacity** on the compositor. Rejected: multiple canvases (hero + background) — duplicate GPU cost and two copies of the mark on screen; CSS-3D image tricks — no real depth or lighting.

3.1 Logo fidelity

`public/logo.jpeg` is raster on black, so the pipeline is: knock out the background → `public/logo-mark.png` (transparent, cropped) → **auto-trace with potrace** into two smooth vector path sets (navy N, silver T) stored in `src/components/logo-shapes.ts`. The trace was visually verified against the original; regenerate it if the brand mark ever changes.

4 Architecture

Unit	Responsibility
<code>components/ambient-background.tsx</code>	The fixed inset-0 -z-10 pointer-events-none layer, aria-hidden. Owns mouse parallax and visibility rules; returns null on /admin*. Lazy-loads the 3D scene.
<code>components/ambient-logo-3d.tsx</code>	The one WebGL scene: extruded logo (SVGLoader → ExtrudeGeometry, bevels), metallic materials, brand-coloured Lightformer environment, 360° turntable via useFrame + refs. dpr=[1, 1.5], frameLoop="demand" under reduced motion. Disposes geometries on unmount.
<code>components/logo-shapes.ts</code> <code>app/globals.css</code>	The traced vector outlines of the real mark (data only). Ambient tokens (--amb-*), aurora/grid keyframes, .ambient-* classes. Size, opacity, per-theme filter, and parallax live on the CSS wrapper — the 3D scene stays theme-blind.
<code>app/layout.tsx</code>	Mounts <AmbientBackground /> once, as the first child of <body>.

4.1 Layering (back to front)

1. **Base** — existing --background token (painted by <body>).
2. **Aurora** — three large soft radial-gradient blobs, slowly drifting + breathing (CSS).
3. **Grid** — the faint blueprint grid, full-page, edge-masked (CSS).
4. **3D logo watermark** — centred, min(72vw, 860px) wide, slowly turning; opacity/lift from theme tokens.

The layer sits at -z-10 as a child of <body>: it paints above the body's background fill but below all in-flow content, so opaque bg-card surfaces naturally mask it.

4.2 Mouse parallax

One rAF-throttled pointermove listener writes normalized --amb-mx/--amb-my (-1...1) onto the layer root; each CSS layer translates a different magnitude (grid least, aurora most). Not attached when (hover: none) or prefers-reduced-motion: reduce.

5 Theme behaviour

Token	Dark	Light
--amb-logo-opacity	0.34	0.30
--amb-logo-filter	brightness(1.5) (lift to read on black)	none (true colours)
Aurora colours	indigo/navy, stronger	indigo/blue, softer

The 3D materials never change per theme — only the CSS wrapper's opacity/filter do.

6 Performance & accessibility

- One lazy-loaded WebGL scene site-wide (next/dynamic, ssr:false) — never blocks first paint; dpr capped at 1.5 (the layer is faint; extra resolution is wasted GPU).

- Turntable animates via `useFrame` ref mutation — no React state in the loop.
- Aurora/grid animate only `transform/opacity`; `pointer-events: none`; `aria-hidden`.
- Reduced motion: canvas drops to `frameLoop="demand"` (static frame), CSS animations freeze via the global guard, parallax listener self-disables.
- Manually created geometries are disposed on unmount.

7 Testing

- `ambient-background.test.tsx`: layer renders `aria-hidden` on public pages; the 3D watermark mounts inside its wrapper (scene stubbed under `jsdom`); renders nothing on `/admin` paths.
- Green bar: `npm run build, typecheck, lint, test` all pass.
- Manual: rotation, float, parallax verified in both themes; content readability confirmed; `/admin` suppression confirmed.

8 Build order (as executed)

1. Knock out the logo background → `public/logo-mark.png`; auto-trace → `logo-shapes.ts`.
2. Extruded 3D scene (`ambient-logo-3d.tsx`) with metallic materials + brand lighting.
3. Ambient layer (`ambient-background.tsx`): aurora + grid + 3D watermark + parallax + `/admin` suppression; mounted once in `layout.tsx`.
4. Removed the hero's separate 3D canvas (crystal) — the watermark is the hero visual.
5. Tests + full green bar; visual QA in both themes; tuned opacity/brightness/materials.